

a

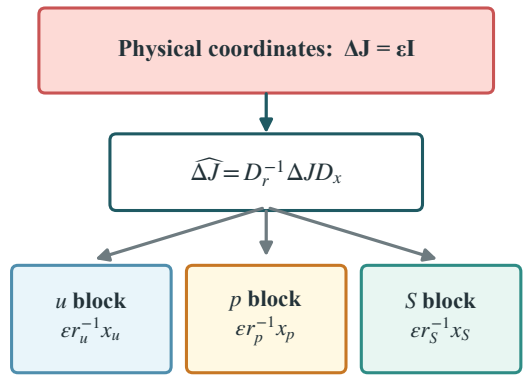
Mixed-field tangent blocks and variable units

	u [m]	p [Pa]	S [-]
displacement residual	J_{uu}	J_{up}	J_{uS}
pressure residual	J_{pu}	J_{pp}	J_{pS}
saturation residual	J_{Su}	J_{Sp}	J_{SS}

Physical-coordinate modification: $\Delta J = \epsilon I$

b

One scalar identity is not one field scale



Equal diagonal entries need not represent equal scaled perturbations.